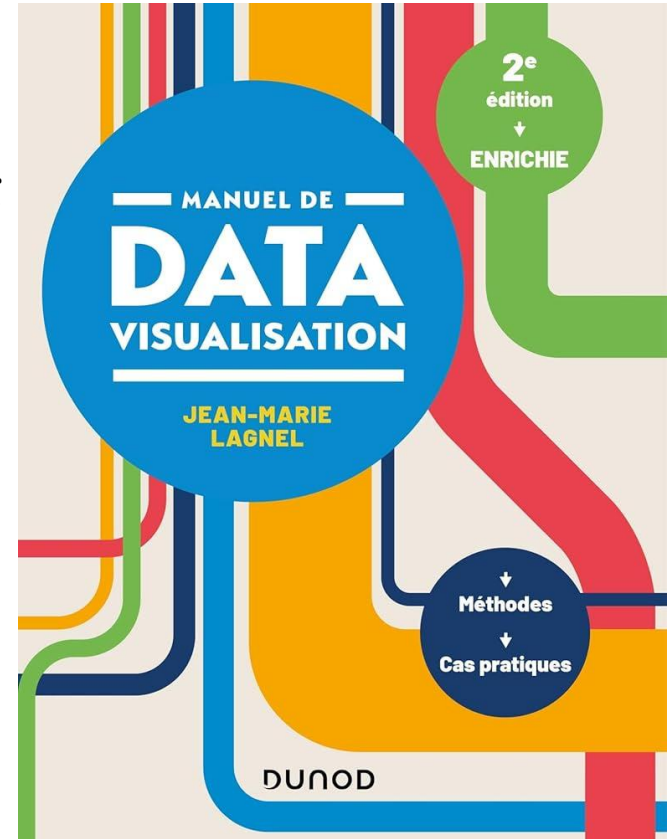


Creating tools to visualize image-based similarities

in AIKON application

What is visualization?

- “Data visualization is used to quickly **present data and make it readable**. It helps to **analyze and facilitate the understanding** of data by effectively **transcribing it into visual form**.”
- “A successful datavisualization **makes a complex subject simple and easy to understand**. It's **generated automatically**, using tools, software or HTML code, and usually consists of a relatively large database.”

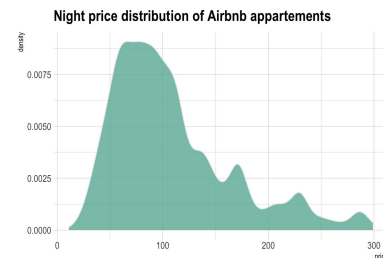
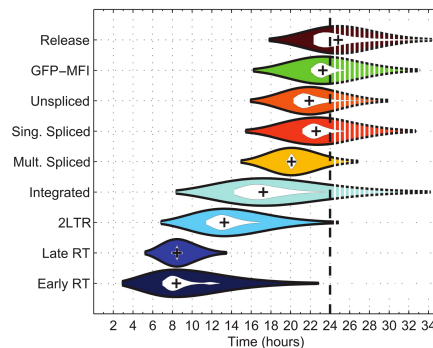
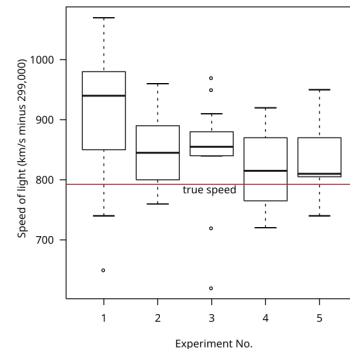
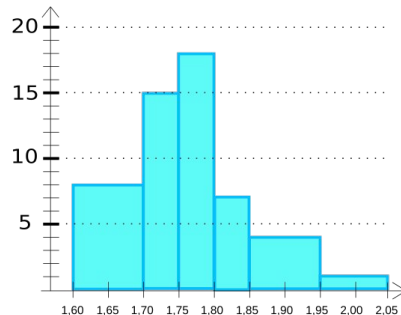


Different types of visualization

Distribution visualizations :

Show how data is spread or distributed

- Histogram
- Box Plot
- Violin Plot
- Density Plot

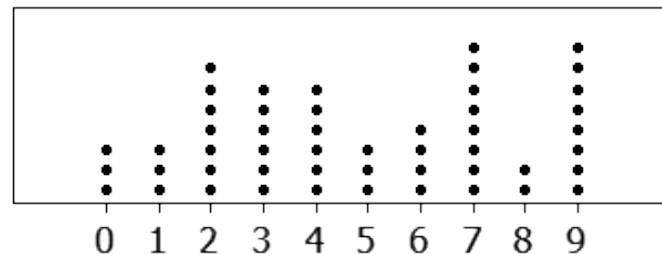
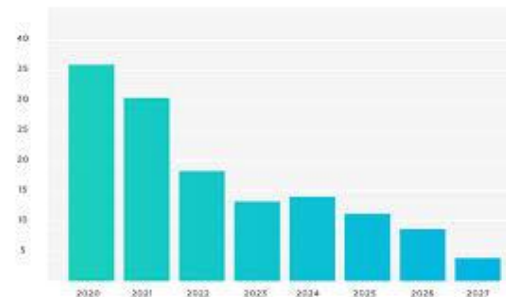


Different types of visualization

Comparison visualizations :

Compare quantities across categories or groups

- Bar chart
- Column chart
- Dot plot

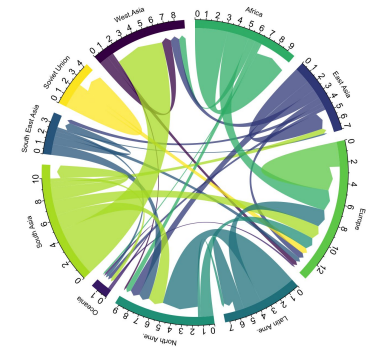
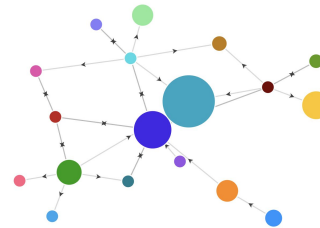
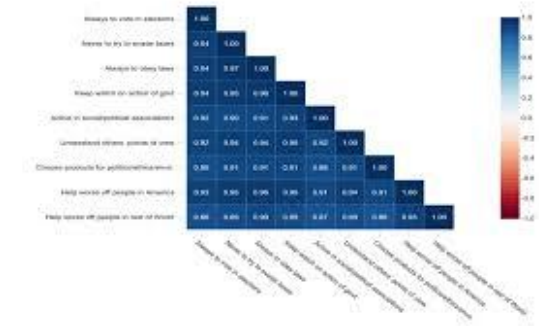
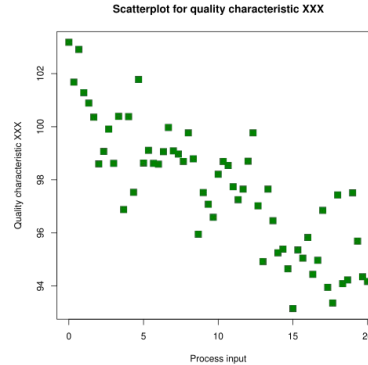


Different types of visualization

Relationship visualizations :

Show correlations or connections between variables or entities

- Scatter plot
- Correlation matrix
- Network graph
- Chord diagram

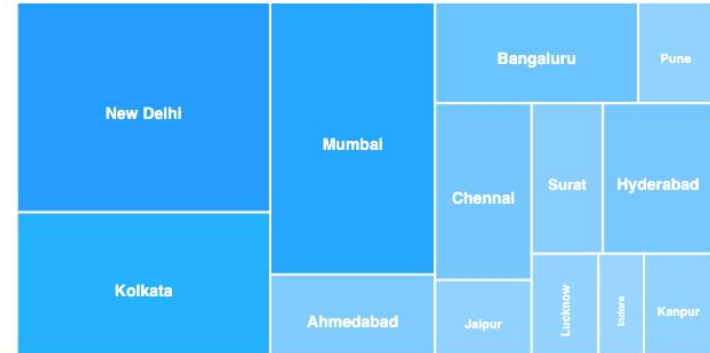
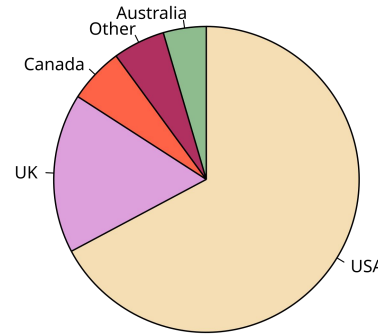


Different types of visualization

Composition Visualizations :

Show parts of a whole

- Pie chart
- Donut chart
- Stacked bar/area chart
- Treemap
- Sunburst chart

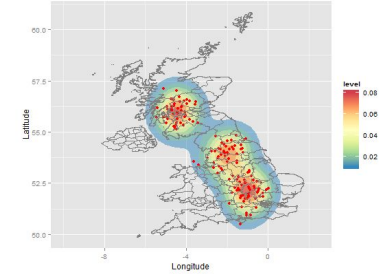
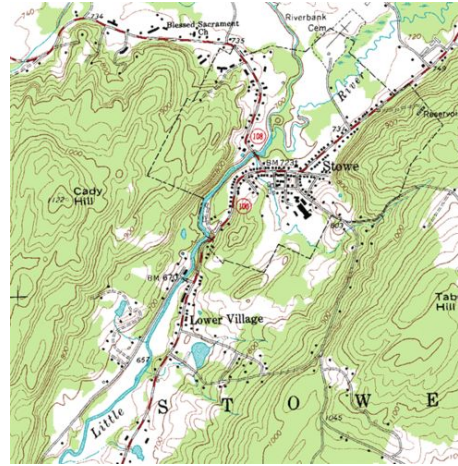


Different types of visualization

Spatial / Geographical Visualizations :

Display data with a spatial or geographic context

- Topographic map
- Flow map
- Heat map

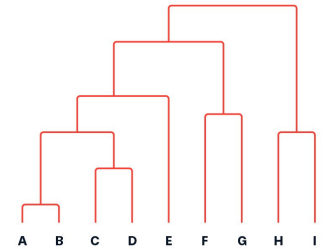
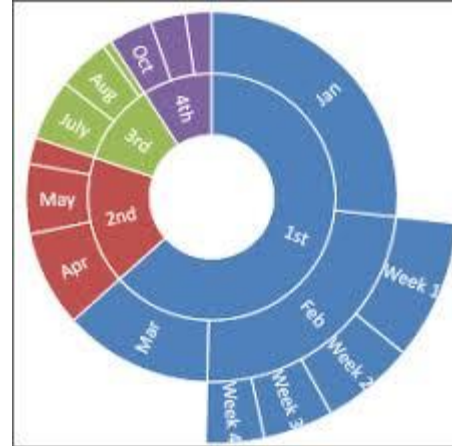
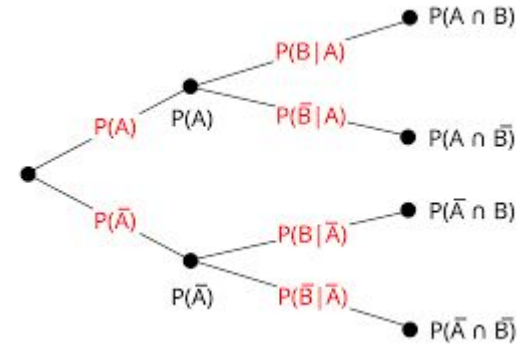


Different types of visualization

Hierarchical visualizations :

Represent nested or tree-like structures

- Tree diagram
- Sunburst
- Dendrogram

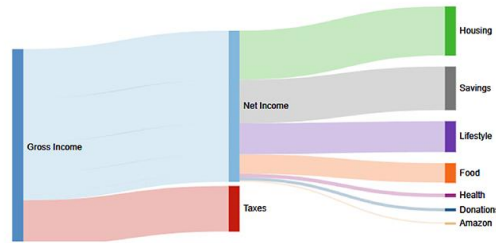


Different types of visualization

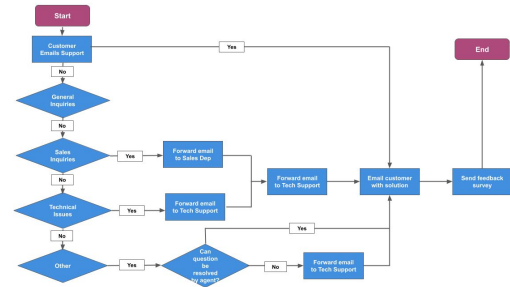
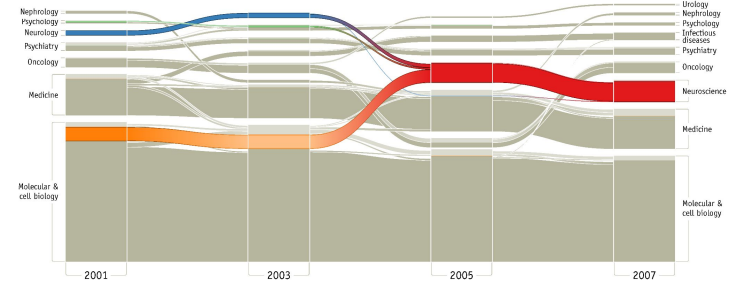
Flow visualizations :

Show movement, processes or paths

- Sankey diagram
- Alluvial diagram
- Flowchart
- Process map



How to Create Sankey Diagram in Excel? Easy Steps

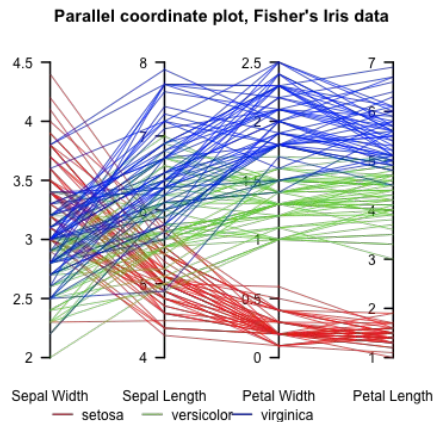
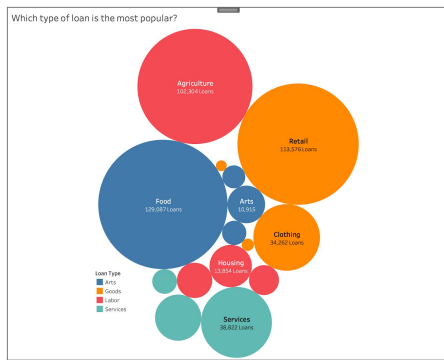
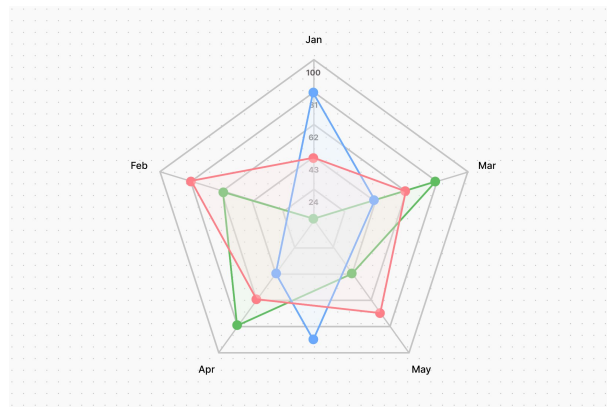


Different types of visualization

Multivariate visualizations :

Display multiple variables in a single chart

- Bubble chart
- Radar/spider chart
- Parallel coordinates



Why create a visualization ?

- The large-scale digitization of cultural content has made it accessible to anyone, anywhere in the world, through web interfaces that serve as libraries or museums.
- The emergence of search engines has radically transformed the way we search for cultural content. In the past, one would start from broad collections (for example, a fonds) and narrow down to the desired object. Today, it is possible to access it directly via a search bar. Visualization can offer a global overview of a collection and allow easy access to individual items.

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1

Visualization of Cultural Heritage Collection Data: State of the Art and Future Challenges

Florian Windhager, Paolo Federico, Günther Schröder, Katrin Glinka, Marian Dörk, Silvia Miksch, Member, IEEE, and Eva Mayr

Abstract—After decades of digitization, large cultural heritage collections have emerged on the web, which contain massive stocks of content from galleries, libraries, archives, and museums. This increase in digital cultural heritage data promises new modes of analysis and increased levels of access for academic scholars and casual users alike. Going beyond the standard representations of search-centric and grid-based interfaces, a multitude of approaches has recently started to enable visual access to cultural collections, and to explore them as complex and comprehensive information spaces by the means of interactive visualizations. In contrast to conventional web interfaces, we witness a widening spectrum of innovative visualization types specially designed for rich collections from the cultural heritage sector. This new class of information visualizations gives rise to a notable diversity of interaction and representation techniques while lending currency and urgency to a discussion about principles such as sovereignty, generality, and critically in connection with visualization design. With this survey, we review information visualization approaches to digital cultural heritage collections and reflect on the state of the art in techniques and design choices. We contextualize our survey with humanist perspectives on the field and point out opportunities for future research.

Index Terms—Information visualization, Introductory and Survey, Digital Libraries, Arts and Humanities

1 INTRODUCTION

Arguably, it is cultural expression and exchange that distinguish humans from other animals. Devising and sharing objects, ideas, and practices enrich behavioral options, facilitates problem-solving, and thus drives the evolution of human collectives [1], [2]. From physical tools and information artifacts to arts and entertainment – cultures create and collect things and pass them on across space and time. While doing so, cultures are changing, and so are the means of transmitting their assets [3]. Digitization has expanded the means for representing and transmitting cultural collections, which makes large stocks of cultural content available, in principle for everyone and everywhere. Against the background of these large data collections, new types of typically web-based interfaces are assuming a role similar to galleries, libraries, archives, and museums as the ‘places’, where cultural heritage (CH) can be experienced [4], [5], [6], [7].

In this report, we collect recent developments of interfaces, which leverage methods of information visualization (InfoVis) to enhance access to cultural collections in order to support their scholarly analysis and causal appreciation. The survey sheds light on this emerging field, and

aims to assess the state of the art for a diverse group of readers and audiences. We assume the findings and discussions to be of relevance for InfoVis researchers and practitioners, cultural scientists and digital humanities scholars, as well as owners, curators and custodians of CH collections. The general purpose of this paper is to explore and consolidate this new field by summarizing recent achievements and by reflecting on future challenges. To do so, we will discuss the background of CH data (sec. 1) and describe our survey methodology (sec. 2). On this basis we introduce the categories of the survey (sec. 3), analyze existing visualization systems (sec. 4), and discuss the findings in relation to a range of contemporary humanities perspectives to derive directions for future research (sec. 5 and sec. 6).

1.1 Concepts of Culture

The concept of ‘culture’ has seen a multitude of definitions [8], [9], [10]. While everyday language often uses ‘culture’ to refer to *artful* things and emphasizes aesthetic or exceptional aspects (“the best that has been thought and known” [11]), many discourses and domains use the term in a much broader and pragmatic way. As such, culture also includes the whole portfolio of *useful* things and thoughts, including the everyday customs and practices that make up how we live as a society “that complex whole which includes knowledge, belief, art, morals, law, custom, and any other capabilities and habits” [12]. From a functionalist perspective, CH thus comprises the whole arsenal of *artful* and *useful* assets that enable and refine collective reproduction. From a more critical perspective, CH objects and contents also always deserve a second look at their implicit functions and motivations (see sec. 5, 3). Seen from such perspectives, CH objects also reveal the functions of social and cultural

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E-mail: isth@inf.tu.wien.ac.at

• Katrin Glinka is with the Chair of Germanic, Neuhäuser Kabinett.
E-mail: k.glinka@uni-ghf.de

• Marian Dörk is with the Urban Complexity Lab, University of Applied Sciences, Potsdam. E-mail: isth@inf.tu.wien.ac.at

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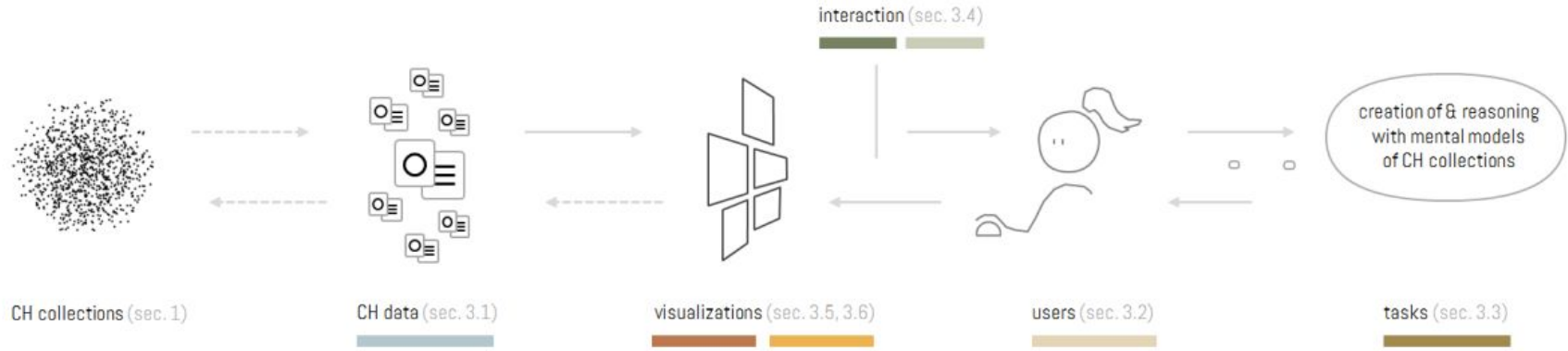


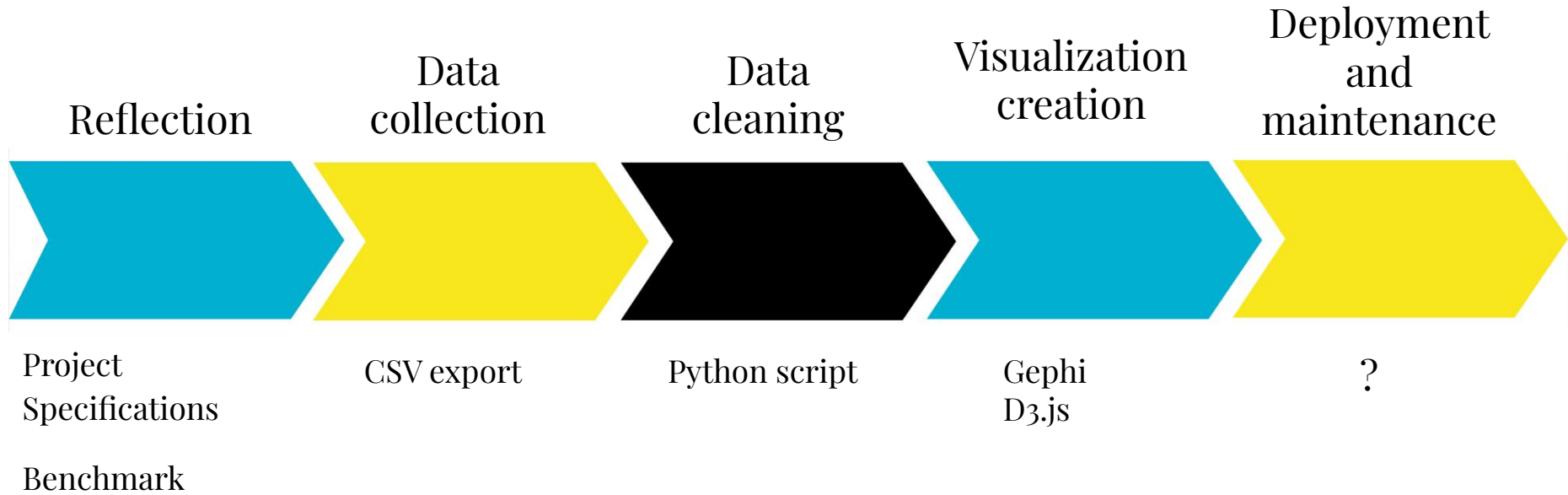
Fig. 3. Schematic lineup of a visualization system in the CH data domain with annotations and colors of the survey's main categories.

F. Windhager et al., "Visualization of Cultural Heritage Collection Data: State of the Art and Future Challenges," in IEEE Transactions on Visualization and Computer Graphics, vol. 25, no. 6, pp. 2311-2330, 1 June 2019, doi: 10.1109/TVCG.2018.2830759.

- Visualization acts as a point of interaction between the user and the data.
- When we make a visualization, we need to pay attention to the type of user we're targeting. There are "expert" and "casual user". "Casual users" need more guidance than "experts" who already know the corpus.

- In the academic context, visualization can play two roles :
 - a demonstrative role : to present scientific knowledge
 - a research role : to make a scientific discoveries by observing a global phenomenon or an irregularity

**How to create a visualization
?**



Reflection phase

Project Specifications :

- Introduction
- Goals and requirements
- Data
- Features
- Prototypes
- Technical restrictions
- Design and application integration
- Deliverables
- Additional resources

Benchmark :

- Introduction
- Methodology
- Presentation of compared solutions
- Recommendations

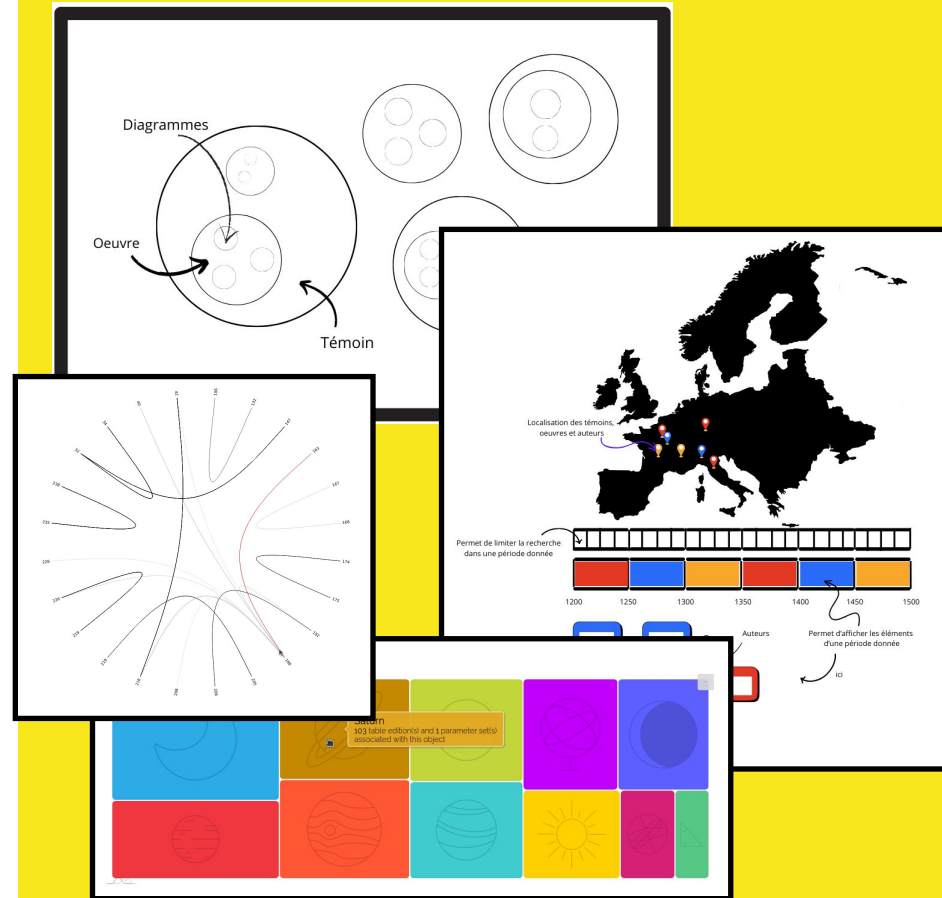
Goals and requirements

Visualize the results of AIKON processing on a given corpus to study the trajectories of the circulation of ideas across culture and over time

- Development of similarity-based visualizations
 - Integration into the AIKON platform for dynamic and automatic updating
 - Designing visual tools for researchers :
 - to use them in conferences to showcase research results
 - to make discoveries within a large dataset
-

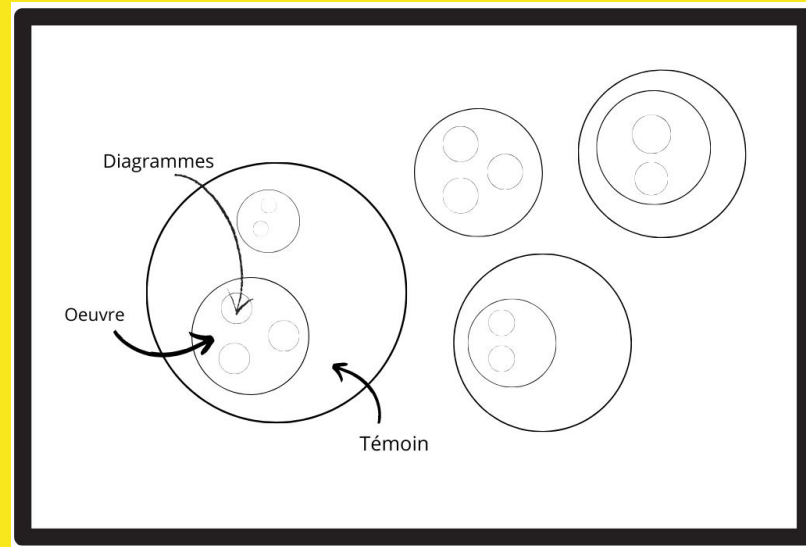
Benchmark

Visualization ideas that were not chosen



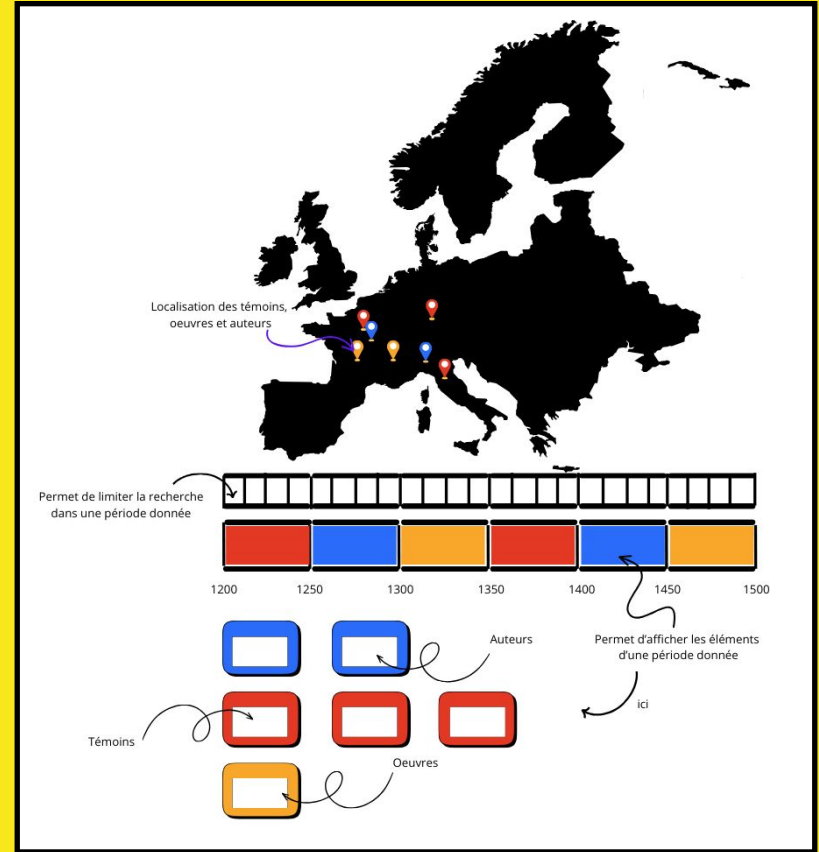
Zoomable circle packing

with [D3.js](#) library



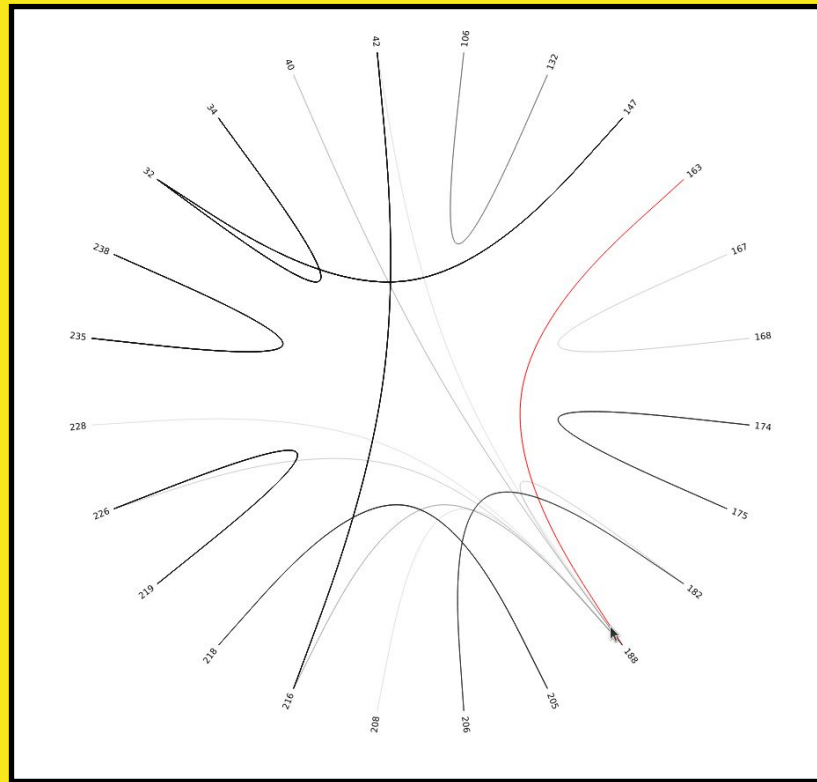
Chronological map

with Leaflet and Timeline JS
libraries



Hierarchical edge bundling

with [D3.js](#) library



Reminder : Data creation in AIKON

EIDA

Admin actions

Manage documents

Computer vision tasks

gr. Z. 303 (coll. 0534)

Witness #350

F1954.34 | National Museum of Asian Art, Smithsonian

Public

sgessner

2025-02-12 17:33

Edit

Add to selection

Work

Zakariyya ibn Muhammad al-Qazwini | Aja'ib al-makhlūqat wa ghara'ib al-mawjudat (The Wonders of Creation and the Oddities of Existence)

Dates

1203-1283

Language

Arabic

Page number

1 ff.

Creation place

Baghdad | IQ

Identification number

F1954.34

Witness #349

None | Unknown place of conservation

mlevy

2024-11-12 23:52

Edit

Add to selection

Work

Unknown author | Jaipur Khasmohor 5292

Dates

-

Language

Sanskrit

Page number

25 pp.

Creation place

-

Identification number

-

Witness #312

Palat. Lat. 1369 | Biblioteca Apostolica Vaticana

Public

ccartier

2024-11-08 11:09

Edit

Add to selection

Work

John of Gmunden | Tractatus Albioni 3

Dates

-

Selection (44)

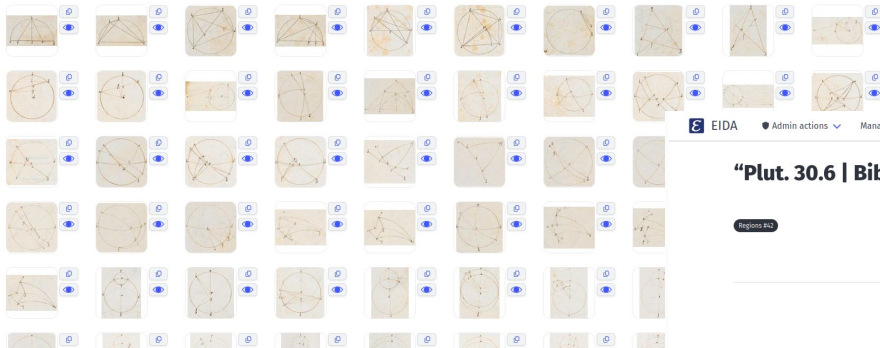
Reminder : Data creation in AIKON - Regions extraction

"Plut. 30.6 | Biblioteca Medicea Laurenziana" regions

Regions 812

Edit Select all Download

All regions Per page Similarity Vectorization



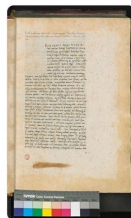
"Plut. 30.6 | Biblioteca Medicea Laurenziana" regions

Regions 812

Edit Select all Download

All regions Per page Similarity Vectorization

1 2 17
1 - 25 / 414



Page 1

Selection (4)

Reminder : Data creation in AIKON - Similarities

EIDA Admin actions Manage documents Computer vision tasks

Regions #42

All regions Per page Similarity Vectorization

Selected regions

Minimal score (0)

0.00025 134.73157

1 2 ... 8

1 - 25 / 186

Page 27

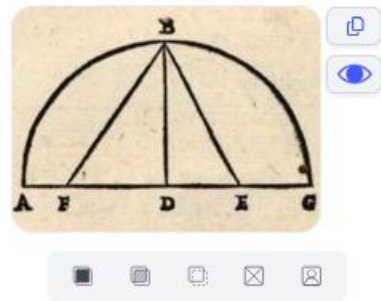
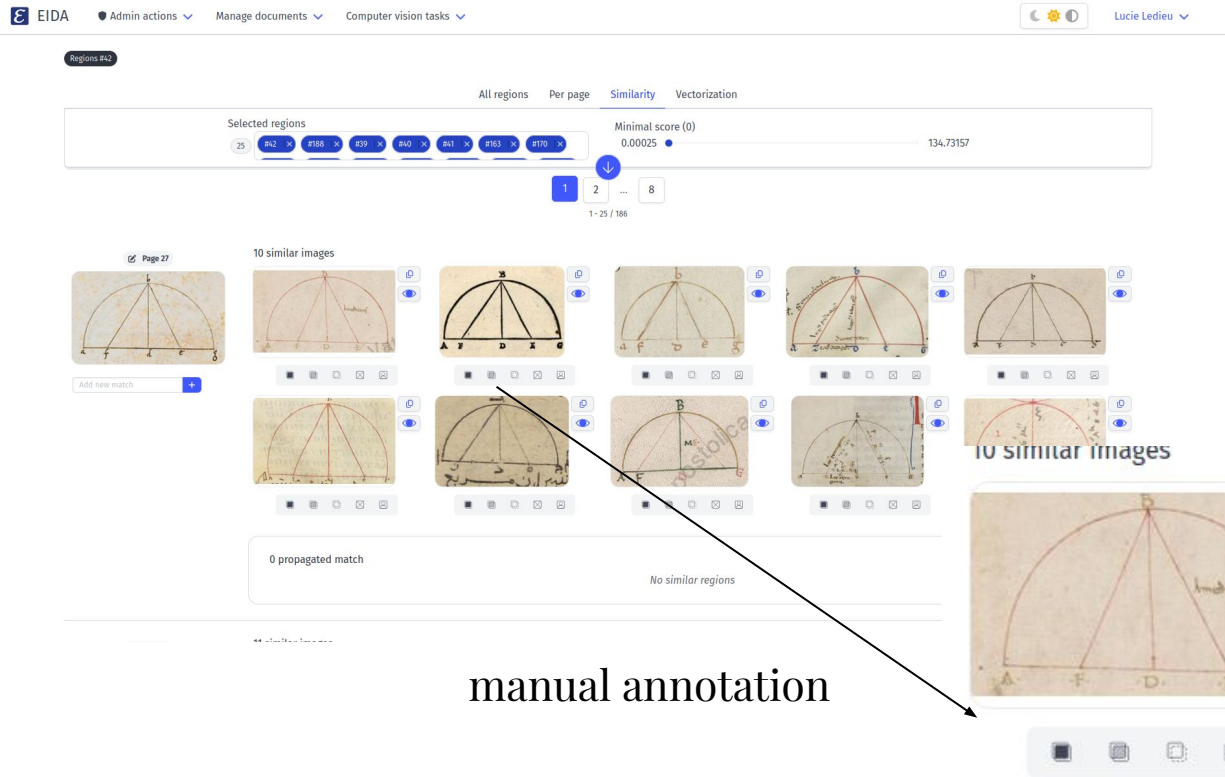
10 similar images

Add new match

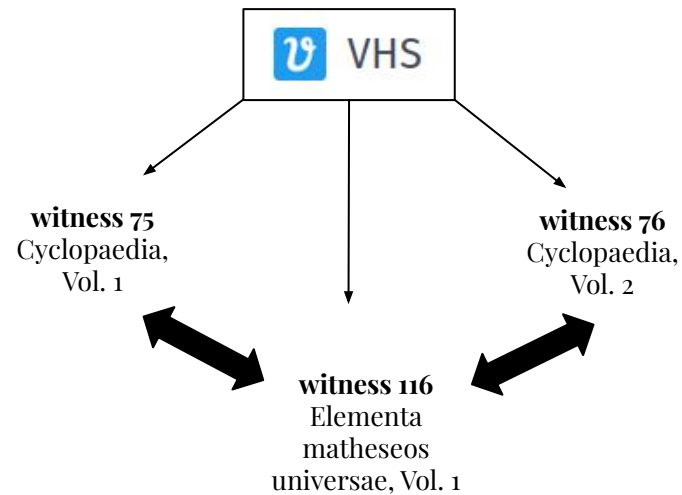
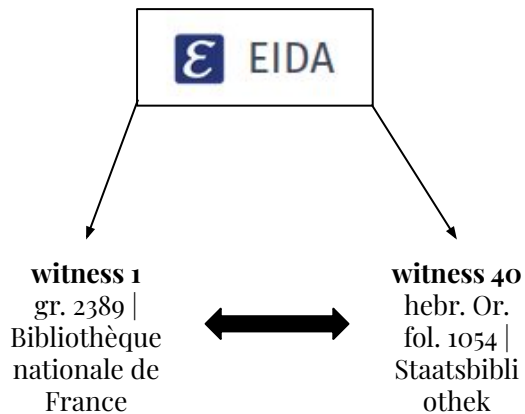
0 propagated match

No similar regions

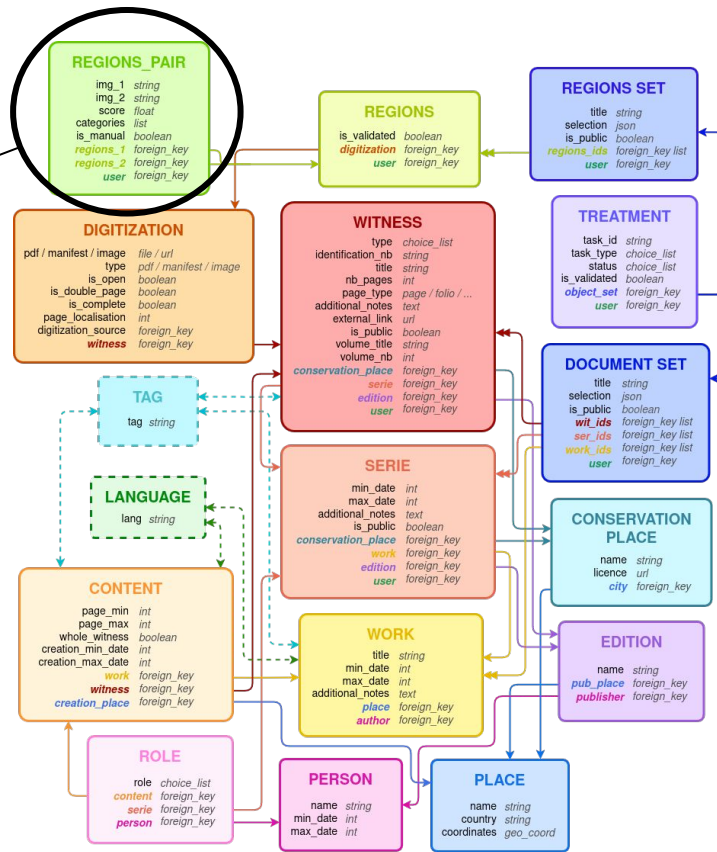
manual annotation



Data collection



region-pairs.csv
csv export
from database



database model

Overview of region_pairs table contents

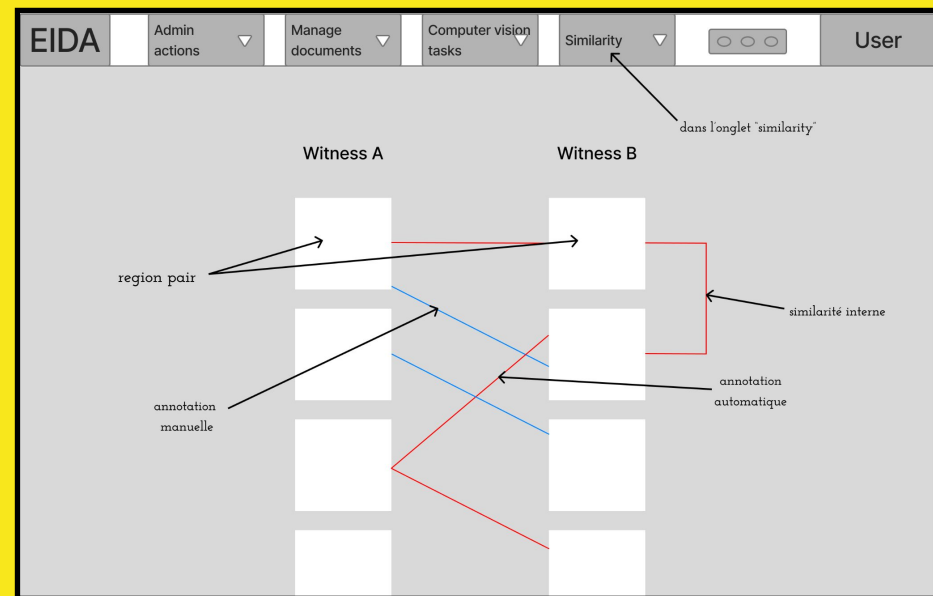
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	id	img_1	img_2	regions_id_1	regions_id_2	category	category_x	created_at	updated_at	is_manual	score	similarity_type	
2		1wit1_man191_0580_1172,1410,429,533.jpg	wit163_img163_061_1249,1715,283,332.jpg	163	188	NULL	{}	2024-04-18 15:02:47.929644+00	2024-04-18 15:02:59.388669+00	False	NULL		1
3		2wit1_man191_0580_1172,1410,429,533.jpg	wit206_man335_0124_457,1606,412,511.jpg	188	216	2	{}	2024-04-23 08:12:39.188658+00	2024-04-23 08:12:39.237295+00	False	NULL		1
4		3wit1_man191_0580_1172,1410,429,533.jpg	wit206_man335_0402_1189,358,391,473.jpg	188	216	1	{}	2024-04-23 08:12:46.359862+00	2024-04-23 08:12:46.439726+00	False	NULL		1
5		4wit1_man191_0579_264,1097,467,569.jpg	wit201_man213_0056_109,551,434,481.jpg	188	208	NULL	{}	2024-04-23 08:13:09.611264+00	2024-04-23 08:13:10.746924+00	False	NULL		1
6		5wit1_man191_0664_472,905,489,600.jpg	wit40_img40_0103_1093,289,660,790.jpg	188	40	NULL	{}	2024-04-23 08:13:19.344236+00	2024-04-23 08:13:25.866735+00	False	NULL		1
7		6wit32_img32_0114_125,242,459,634.jpg	wit34_img34_0118_1763,1836,3675,3884.jpg	32	34	4	{}	2024-04-26 10:31:43.942884+00	2024-04-26 10:31:43.979684+00	False	NULL		1
8		7wit32_img32_0114_125,242,459,634.jpg	wit147_img147_0205_209,244,1052,1349.jpg	147	32	4	{}	2024-04-26 10:31:48.992251+00	2024-04-26 10:31:49.036081+00	False	NULL		1
9		8wit32_img32_0114_125,242,459,634.jpg	wit147_img147_0214_453,955,708,620.jpg	147	32	4	{}	2024-04-26 10:31:50.736316+00	2024-04-26 10:31:50.750631+00	False	NULL		1
10		9wit32_img32_0114_125,242,459,634.jpg	wit34_img34_0139_321,5010,3606,2477.jpg	32	34	4	{}	2024-04-26 10:31:52.94523+00	2024-04-26 10:31:52.979871+00	False	NULL		1
11		10wit32_img32_0114_125,242,459,634.jpg	wit34_img34_0049_283,1309,3824,4889.jpg	32	34	4	{}	2024-04-26 10:31:59.510784+00	2024-04-26 10:31:59.563555+00	False	NULL		1
12		11wit32_img32_0114_125,242,459,634.jpg	wit34_img34_0124_3006,1351,2312,2310.jpg	32	34	4	{}	2024-04-26 10:32:00.609474+00	2024-04-26 10:32:00.654651+00	False	NULL		1

The columns that interest us :

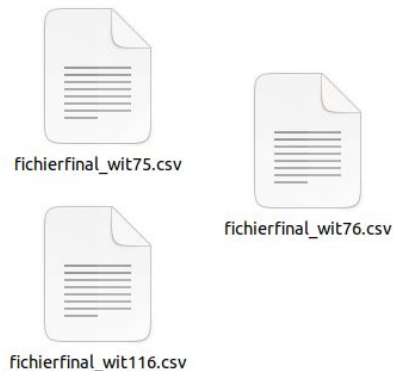
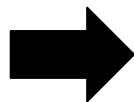
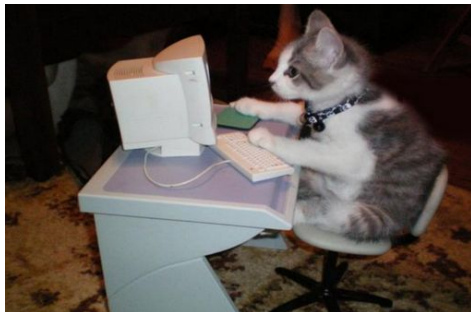
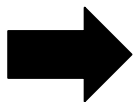
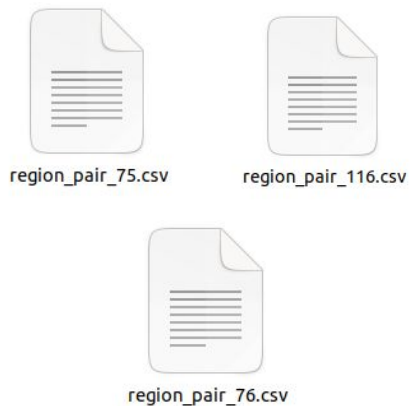
- id : unique identifier for each image pair
- img_1 : region 1 file
- img_2 : region 1 file
- score : similarity score
- similarity_type :
 - 1 : exact match
 - 2 : partial match
 - 3 : semantic match
 - 4 : no match

Visualization 1

Bipartite graph
VHS



Data cleaning



```
import pandas as pd
import re

pd.read_csv('region_pair_75.csv', index_col=[0])

# ... (code for reading and cleaning data) ...

# ... (code for saving cleaned data) ...
```

Data cleaning



- Keep only the lines containing the witnesses we are interested in the img_1 and img_2 : wit116, wit75, wit76
- Delete columns that are not interesting : regions_id_1, regions_id_2, created_at, updated_at, category, category_x, is_manual
- Delete similarity types that are not interesting : 3 (semantic match) and 4 (no match)
- Delete internal links

don't forget the folder with the images of the regions in it to be integrated into the visualization !!!!

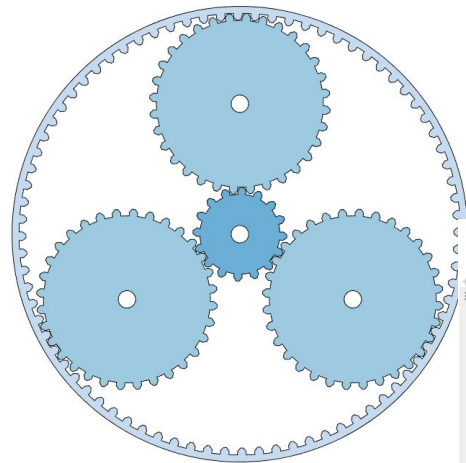
Examples of d3.js use

https://observablehq.com/@d3/gallery?utm_source=d3js-org&utm_medium=hero&utm_campaign=trv-observable

Epicyclic gearing

From Wikipedia: "one or more outer gears, or planet gears, revolving about a central, or sun gear... Epicyclic gearing systems also incorporate the use of an outer ring gear or annulus, which meshes with the planet gears."

Fixed gear ☐ annulus ☒ planets ☐ sun



Tadpoles

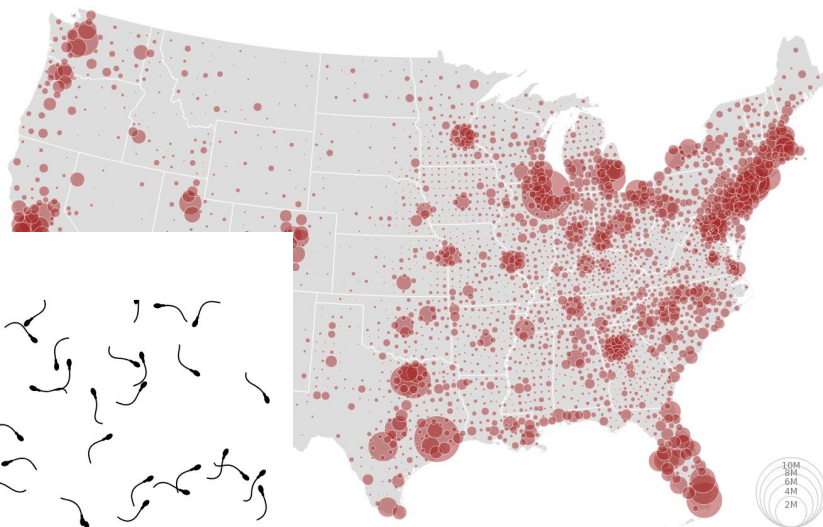
A recreation of the [Paper.js example](#) (though lacking the playful flocking behavior!).



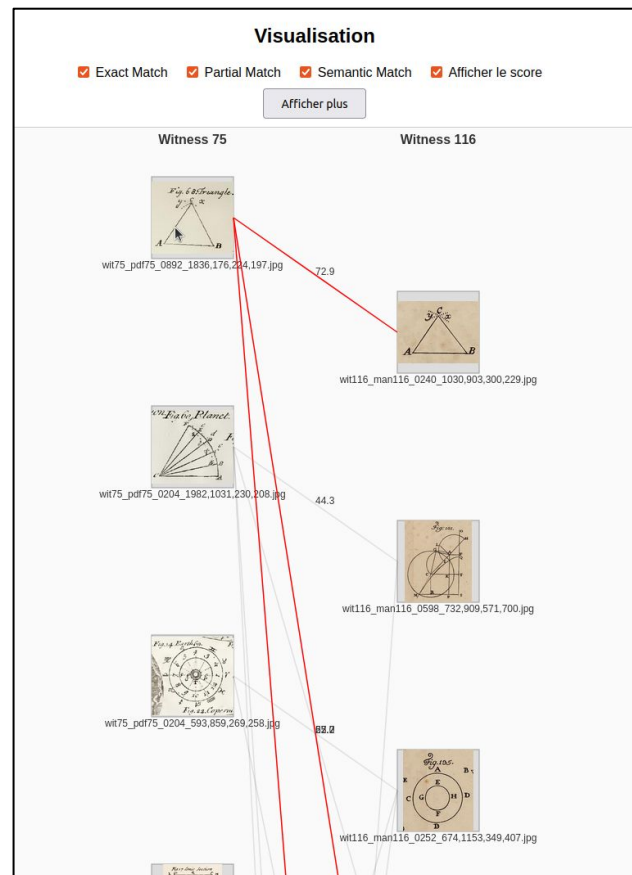
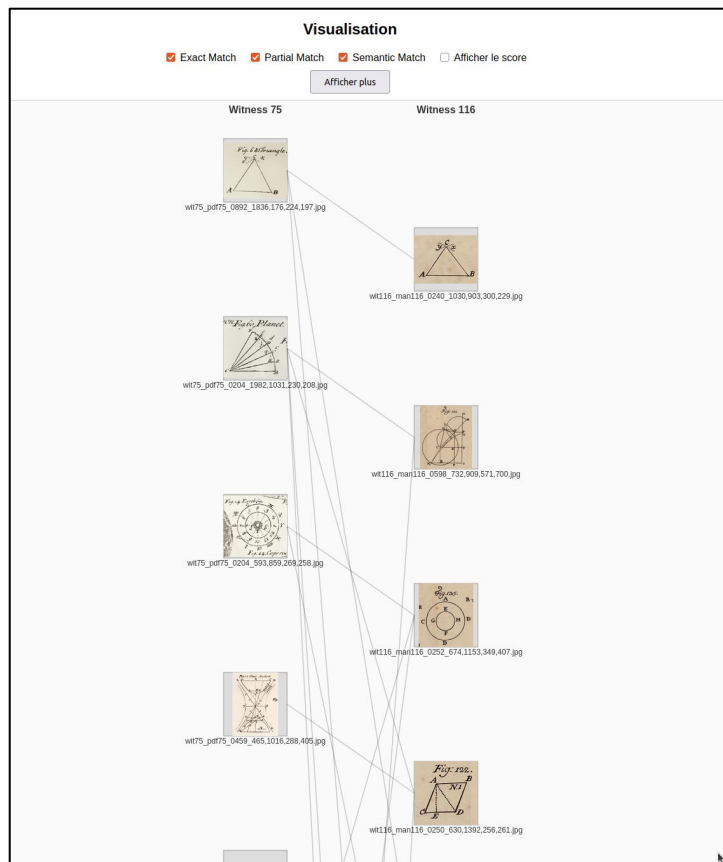
D3 > GALLERY

Bubble map

Estimated population by county, 2016. See also the [spike map](#) as an alternative presentation of this data. Data: [American Community Survey](#)

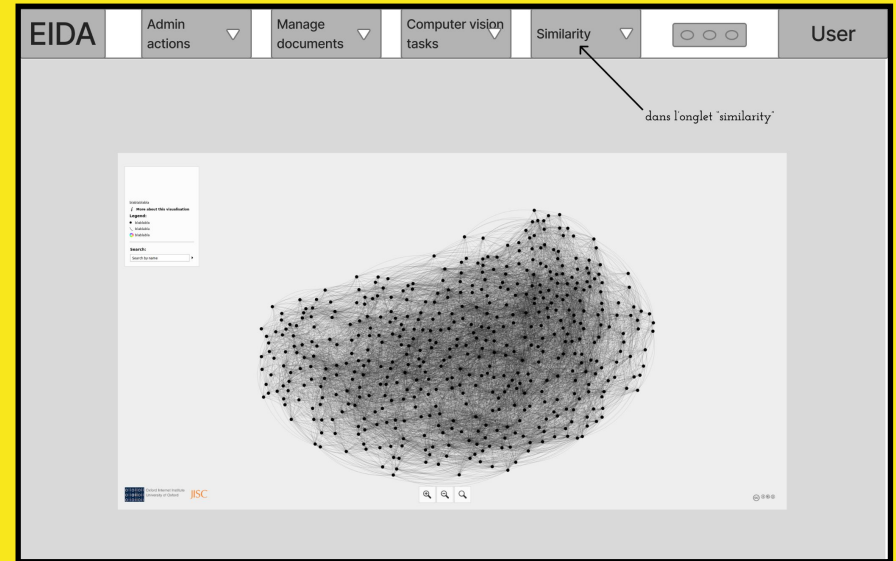


Final visualization



Visualization 2

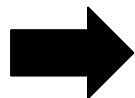
Force-directed layout
EIDA



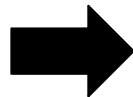
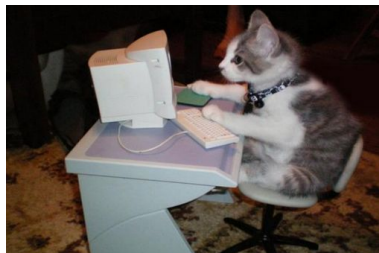
Workflow



region-pairs.csv



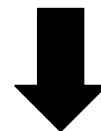
data cleaning



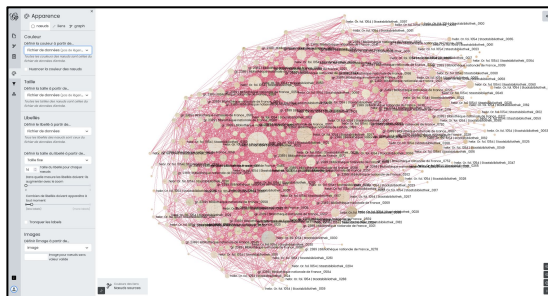
nodes_gephi.csv



edges_gephi.csv



gephi-o.9.6



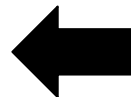
gephi lite



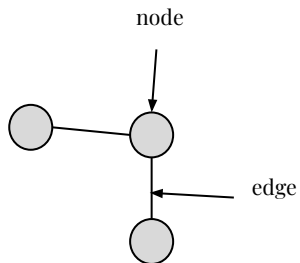
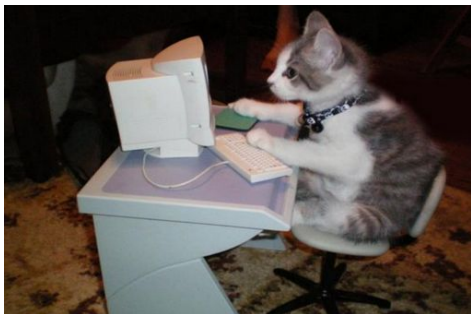
data cleaning



visualisation.gexf



Data cleaning



- keep only the region pairs of our two witnesses (wit40 and wit1)
- delete lines in which there are not only relationships between our two witnesses
- delete columns that do not interest us (regions_id_1, regions_id_2, created_at, updated_at)
- to use Gephi, we need to create two csv files : one for nodes and one for edges

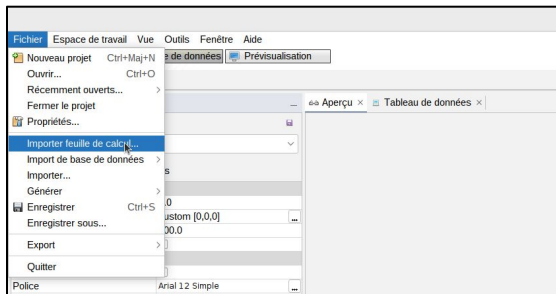


Gephi

- Free and open source software for network graph visualization and analysis
- To represent relational data as graphs composed of nodes and links
- Main functions :
 - automatic graph layout with algorithms
 - network statistics
 - visual customization (sizes, colors, labels)
 - export in image or graph format
- documentation : <https://gephi.org/users/>



Using Gephi

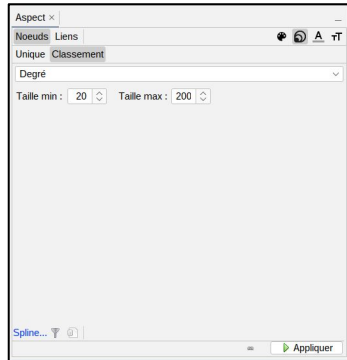


Import the two csv
files (nodes and edges)

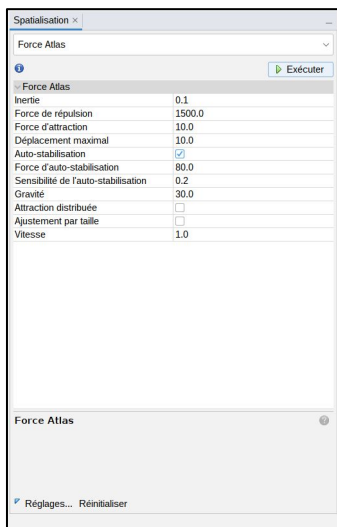
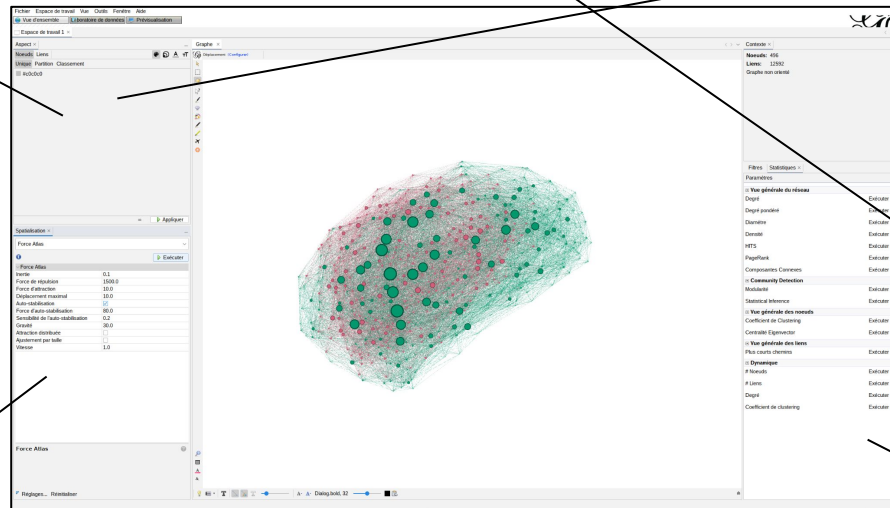
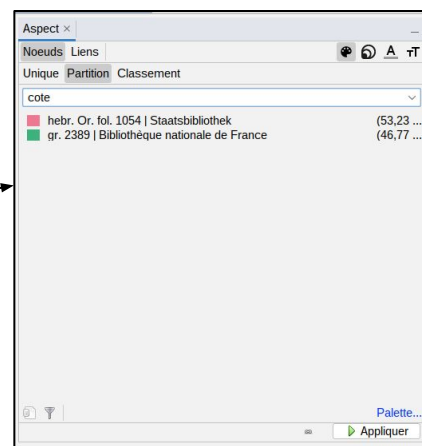
The screenshot shows the 'Tableau de données' window in Gephi. The window displays a list of data sources, including 'region' and 'cote'. The 'region' column lists various regions, and the 'cote' column lists various coastal areas. The table is organized into columns: 'id', 'Label', 'Interval', 'region', and 'cote'. The data is presented in a tabular format, with rows corresponding to different data entries.

	id	Label	Interval	region	cote
1	gr_2389	Bibliothèque nationale de France_0664		wet1_man191_0664_472.905.489.600.jpg	gr_2389
2	gr_2389	Bibliothèque nationale de France_0596		wet1_man191_0596_464.767.485.433.jpg	gr_2389
3	gr_2389	Bibliothèque nationale de France_0032		wet1_man191_0032_295.1317.583.353.jpg	gr_2389
4	gr_2389	Bibliothèque nationale de France_0009		wet1_man191_0009_166.1325.578.516.jpg	gr_2389
5	gr_2389	Bibliothèque nationale de France_0027		wet1_man191_0027_1143.2063.303.245.jpg	gr_2389
6	gr_2389	Bibliothèque nationale de France_0031		wet1_man191_0031_857.2013.543.341.jpg	gr_2389
7	gr_2389	Bibliothèque nationale de France_0034		wet1_man191_0034_320.1234.564.556.jpg	gr_2389
8	gr_2389	Bibliothèque nationale de France_0035		wet1_man191_0035_206.413.593.326.jpg	gr_2389
9	gr_2389	Bibliothèque nationale de France_0035		wet1_man191_0035_508.1914.535.313.jpg	gr_2389
10	gr_2389	Bibliothèque nationale de France_0036		wet1_man191_0036_975.1114.897.668.jpg	gr_2389
11	gr_2389	Bibliothèque nationale de France_0038		wet1_man191_0038_1053.1397.549.588.jpg	gr_2389
12	gr_2389	Bibliothèque nationale de France_0039		wet1_man191_0039_881.987.599.603.jpg	gr_2389
13	gr_2389	Bibliothèque nationale de France_0040		wet1_man191_0040_1134.1699.182.196.jpg	gr_2389
14	gr_2389	Bibliothèque nationale de France_0048		wet1_man191_0048_1089.1898.515.402.jpg	gr_2389
15	gr_2389	Bibliothèque nationale de France_0049		wet1_man191_0049_269.1891.486.415.jpg	gr_2389
16	gr_2389	Bibliothèque nationale de France_0050		wet1_man191_0050_1079.815.449.424.jpg	gr_2389
17	gr_2389	Bibliothèque nationale de France_0050		wet1_man191_0050_339.417.488.446.jpg	gr_2389
18	gr_2389	Bibliothèque nationale de France_0051		wet1_man191_0051_103.1074.635.400.jpg	gr_2389
19	gr_2389	Bibliothèque nationale de France_0051		wet1_man191_0051_811.1094.777.365.jpg	gr_2389
20	gr_2389	Bibliothèque nationale de France_0052		wet1_man191_0052_861.1929.706.412.jpg	gr_2389
21	gr_2389	Bibliothèque nationale de France_0054		wet1_man191_0054_376.1510.554.543.jpg	gr_2389
22	gr_2389	Bibliothèque nationale de France_0058		wet1_man191_0058_429.553.429.407.jpg	gr_2389
23	gr_2389	Bibliothèque nationale de France_0064		wet1_man191_0064_403.1238.451.420.jpg	gr_2389

Their contents can be viewed and
modified in the “Data laboratory”.



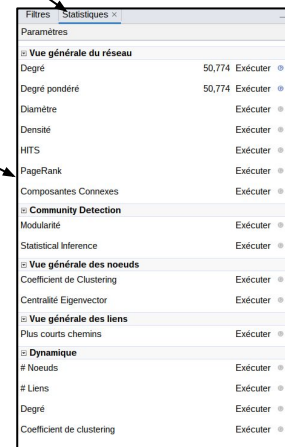
Aspect : to customize the visual appearance of nodes and edges according to associated data (e.g. shelfmark) or calculated statistics



Spatialization : to automatically position graph nodes in space using a spatialization algorithm (Force Atlas, Fruchterman-Reingold, Yifan Hu...)

Overview

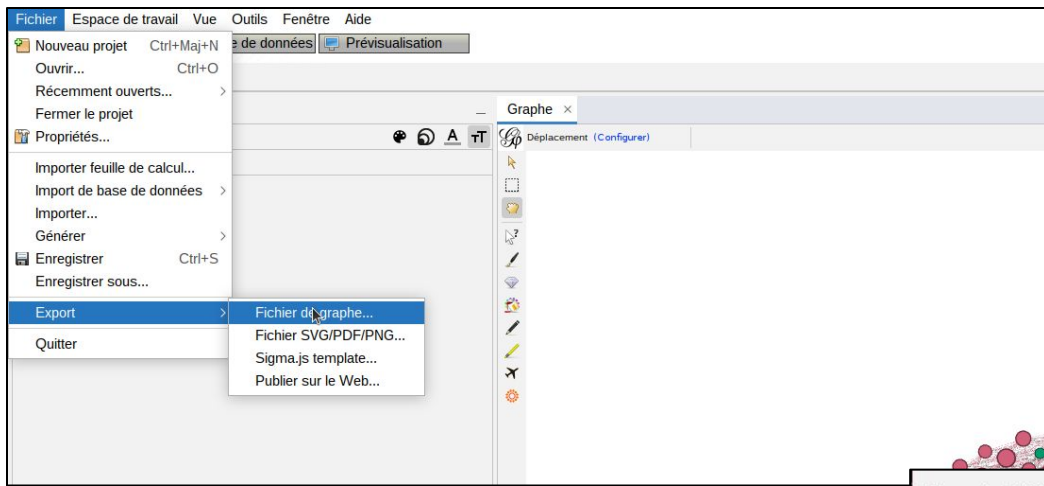
Statistics : to run quantitative analyses on your graph to measure its structural properties



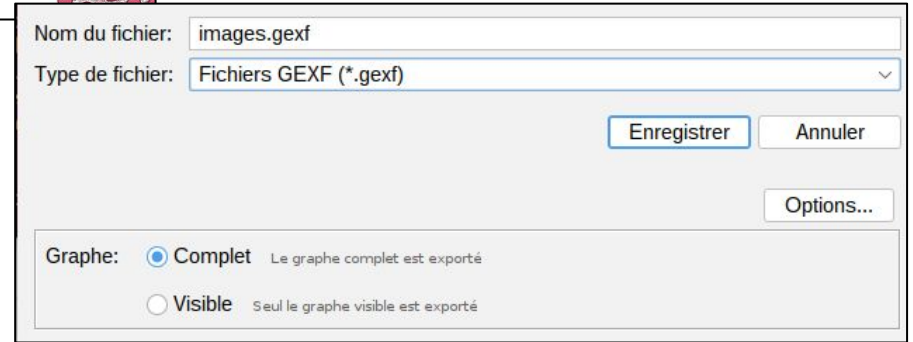


Preview

Generate a clean, exportable rendering of the graph by adjusting the final styling parameters (font, color, etc.).



A gexf file (Graph Exchange XML Format) is a format for storing graph data (nodes, links) and enabling the exchange and visualization of networks in different software packages.



Export

Data cleaning



Unfortunately, it's not possible to insert images into nodes on classic Gephi: we have to use Gephi Lite online.

To finish the visualization, we need to perform a final data clean-up to add the URL to the images in the EIDA database to our gexf file (we can't use images stored locally on our computer).



Gephi Lite

 Apparence

☐ nœuds / ☐ liens ☒ graph

Couleur

Définir la couleur à partir de...

Fichier de données (pos de légende...)

Toutes les couleurs des nœuds sont celles du fichier de données d'entrée.

☐ Nuancer la couleur des nœuds

Taille

Définir la taille à partir de...

Fichier de données (pos de légende...)

Toutes les tailles des nœuds sont celles du fichier de données d'entrée.

Libellés

Définir le libellé à partir de...

Fichier de données

Tous les libellés des nœuds sont ceux du fichier de données d'entrée.

Libellés

Définir le libellé à partir de...

Fichier de données

Tous les libellés des nœuds sont ceux du fichier de données d'entrée.

Définir la taille du libellé à partir de...

Taille fixe

10 Taille du libellé pour chaque nœuds

Dans quelle mesure les libellés doivent-ils augmenter avec le zoom

0 1

Combien de libellés doivent apparaître à tout moment

(less labels) (more labels)

☐ Trouver les labels

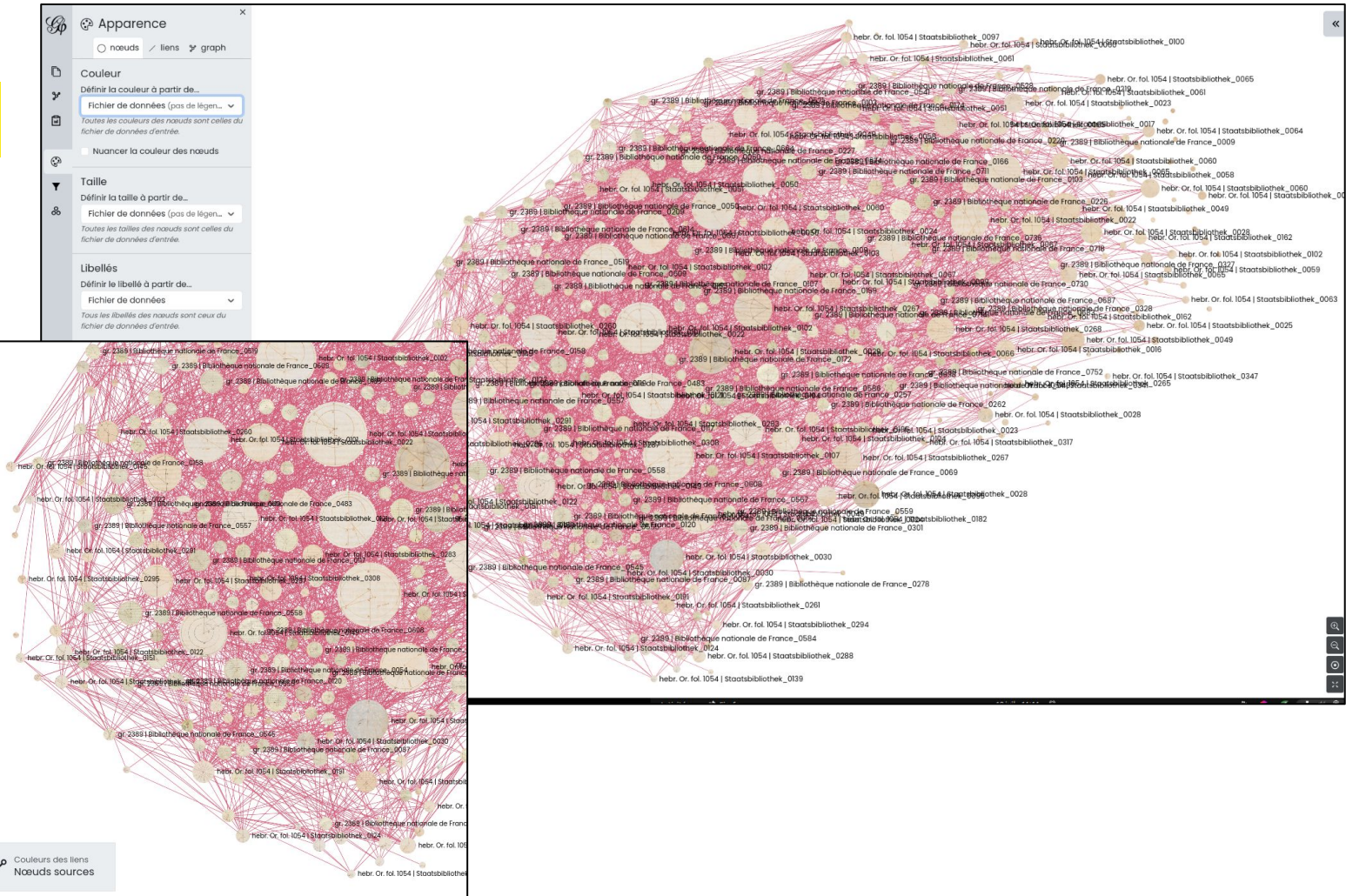
Images

Définir l'image à partir de...

Image

Image pour nœuds sans valeur valide

Couleurs des liens
Nœuds sources



**What will happen to these
visualizations after the
internship ?**

**It's a little
complicated...**

We can't integrate them into the
application just like that !



Problems

- The main problem for bipartite visualization is that it is not connected to the database
- The main problem with force-directed layout is that it is designed to be viewed on Gephi Lite online
- In both cases, since we're talking about a proof of concept, it's not possible to choose (via a drop-down list, for example) the witnesses we want to visualize

Solutions

- For the Gephi visualization, we will take our inspiration from the visualization created during the internship, but code a new visualization with Javascript libraries that will be connected to the database to integrate it into AIKON.
- For bipartite visualization this should be easier, as all you need to do is integrate the code into the application and connect the database. We'll need to add a few functionalities, such as drop-down menus for selecting witnesses.

"While the proverb 'a picture is worth a thousand words', often attributed to Confucius, seems to be supported by recent developments in data visualization, it is worth recalling that without critical discourse, such visualization lacks full narrative or explanatory legitimacy. But if the image does indeed speak — whether to illustrate a scholarly article or to guide a researcher in their inquiry — it is probably because visualization acts as a language that reaches its reader in an almost organic, visceral, and instinctive way, establishing a new relationship with the data. [...] Let us therefore advocate for historians not to outsource the technical components of their research, but to engage directly with data visualization themselves, as part of an exploratory and methodological approach, realizing that mapping their data is already a way to create new knowledge."

Grandjean, Martin, « *Introduction à la visualisation de données : l'analyse de réseau en histoire* », dans *La visualisation des données en histoire*, Lausanne, Éditions Chronos, 2015, p.127.



Bibliography

- **Grandjean, Martin**, « *Introduction à la visualisation de données : l'analyse de réseau en histoire* », dans *La visualisation des données en histoire*, Lausanne, Éditions Chronos, 2015, p.127.
- **Lagnel, Jean-Marie**, *Manuel de datavisualization*, Paris, Dunod, 2001.
- **F. Windhager et al.**, "Visualization of Cultural Heritage Collection Data: State of the Art and Future Challenges," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 6, pp. 2311-2330, 1 June 2019.

**Thank you for your
attention**



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